

Jayant Toleti

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EXPERIENCE

AI Engineering Intern

April 2024 – June 2024

Tecnotree Corporation

Hyderabad, India

- Developed a RAG Evaluation Pipeline using MLFlow and LangChain to evaluate an existing RAG System
- Increased recommendation accuracy by 38% by implementing a Collaborative Filtering approach.
- Investigated the potential of incorporating online Reinforcement Learning (RL) techniques for dynamic recommendation generation.

EDUCATION

Amrita Vishwa Vidyapeetham

8.75/10

Bachelors of Technology in Computer Science and Engineering (Artificial Intelligence)

2021 - Pursuing

PROJECTS

Crack Detection and Segmentation for Infrastructural Analysis | *Python, PyTorch*

04/2024

- * Collected over 11,200 images merged from available crack segmentation datasets.
- * Deployed a Custom U-Net with a VGG 16 Architecture for Segmentation Task.
- * Achieved a remarkable mean IoU of 82% and a 28% reduction in computation time.

Towards Fluent Expression: The Impact of Deep Learning in Grammar Correction | *Python, NLP*

02/2024

- * Processed over 10,00,000 Sentences in the Lang-8 Corpus and performed Grammatical Error Correction.
- * Leveraged FastText Embeddings and created 4 Encoder-Decoder models and a T-5 Transformer, then evaluated the results.
- * Achieved a remarkable GLEU score of 86% while reducing training time by 51%.

Drug Cardio-Toxicity Analysis By Utilising Graph Neural Networks | *Python, TF-GNN*

12/2023

- * Developed a custom Graph Neural Network to evaluate the toxicity of drugs.
- * Built 5,000 GraphTensors for each data point present in the dataset.
- * Implemented Various Architectures to select the best model with a validation accuracy of 90.60%.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS, R and MATLAB.

Developer Tools: Git, Google Cloud Platform, VS Code, PyCharm, IntelliJ, Eclipse.

Others: Arduino, Raspberry Pi, Figma, 3D CAD (Fusion 360), ROS.

ACHIEVEMENTS

- **Published Research** on Radio Signal Classification using DL methods in the 14th ICCNT Conference
- Engaged in **Amazon ML Summer School 3rd Edition** in 2023.
- Finished **3rd** out of 2000 in the **Intel Unnati Rapid Summer Challenge 2022**, leading a team to overcome complex challenges in hosting multiple Tenants on FPGAs, resulting in breakthrough solutions and recognition.

CERTIFICATES AND COURSES

- People and Soft Skills for Professional and Personal Success. | 09/2023
- J.P. Morgan & Chase Software Engineering Virtual Experience. | 05/2023
- R Programming A-Z™: R For Data Science With Real Exercises! | 01/2023